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SUMMARY

- Principal-level engineer who builds AI and machine learning systems end to end: agentic LLM pipelines, classical classifiers, and the data platforms that feed both.
- Hands-on with modern GenAI — self-hosted LLMs, agentic tool-calling, MCP servers, RAG, prompt engineering, model benchmarking, GPU/VRAM scheduling — on top of 20+ years of production ML.
- Designed or implemented 50+ production classifiers and owned a portfolio of ~200 for McAfee/Intel Global Threat Intelligence; two patents in reputation and malicious-traffic detection.
- Works from evidence: benchmark harnesses, held-out evaluation, and measured routing decisions instead of vendor claims. Technical leader, instructor, and frequent conference speaker.

CORE STRENGTHS

- Agentic AI & LLM systems: local inference, tool-calling, MCP, RAG, prompt and context engineering
- Machine learning: feature engineering, classifier design, ensembles, graph methods, model evaluation
- Data analysis at scale: telemetry mining, anomaly surfacing, labeling strategy, drift and quality monitoring
- ML platforms: streaming and batch pipelines, lakehouse, low-latency edge model serving
- AI evaluation: reproducible benchmarks, latency/throughput profiling, failure-mode analysis, output safety
- Technical leadership: architecture ownership, mentoring, cross-functional and executive alignment

TECHNICAL SKILLS

AI / LLM: AWS Bedrock, Ollama, llama.cpp, Qwen3-Coder, Hermes, Claude / Claude Code, ChatGPT, ZeroClaw, MCP (JSON-RPC tool servers), agentic orchestration, RAG, prompt engineering, KV-cache quantization, GPU/VRAM scheduling, whisper.cpp (ASR), Piper (TTS), openWakeWord

Machine Learning & Analytics: Graph Neural Networks, Bayesian networks, SVM, Random Forest, XGBoost, clustering and anomaly detection, voice embeddings, held-out and A/B evaluation, Spark, Map-Reduce, Hadoop, SageMaker, Jupyter, NumPy, ONNX Runtime

Data Platform: ETL/ELT design, Iceberg, Glue, Flink, EMR, QuickSight, telemetry analytics, feature extraction, data quality and labeling; Postgres, MySQL, MongoDB, OpenSearch, Greenplum, Neo4j, Redis

Cloud & Infrastructure: AWS (EC2, S3, Lambda, Aurora, DynamoDB, CloudWatch, IAM, Route 53, NLB/ALB, CloudFormation), OCI, Kubernetes, Docker, Helm, Terraform, CI/CD, GitHub Actions, systemd; Python, Bash, SQL; Linux, BSD

Security Domain: email/IP/domain reputation, malware and mobile app analysis, intrusion detection, protocol analysis, zero trust, MITRE ATT&CK, NIST, GDPR, CAN-SPAM, SBOM

SELECTED AI / ML WORK

HomeAI — Local Agentic Voice Assistant (github.com/adamwoz-personal/homeai)

- Designed and built a fully local, privacy-preserving voice agent: ~21K lines of Python across 30+ modules with 35 test modules and no cloud dependency in the core path.
- Integrated a self-hosted LLM (Qwen3-Coder-30B on Ollama/llama.cpp) with whisper.cpp recognition, Piper synthesis, and openWakeWord detection into one interruptible, barge-in conversational loop; added

speaker recognition from voice embeddings for per-speaker identification.
- Implemented an MCP (JSON-RPC) tool server exposing weather, web research, and home-automation tools to the agent, with dual CLI and HTTP transports carrying distinct risk profiles.
- Wrote a reproducible benchmark suite (time-to-first-token, tokens/sec, KV-cache configurations, model comparison) and used it to raise local generation from ~25 to 131 tok/s by reallocating GPU memory and quantizing the KV cache.
- Authored an evidence-based model-routing guide quantifying the local model's real failure modes — tool-iteration ceilings, malformed tool-call rates, context poisoning — routing work between local and frontier models by measured capability rather than assumption.
- Built the AI safety and privacy layer: utterance validation, response sanitization before speech, and bounded, self-expiring memory with configurable retention and purge.

Applied GenAI in Practice

- Uses agentic coding workflows daily to produce clean, tested, modular code, and has developed a leadership philosophy for AI-centered engineering organizations that treats institutional knowledge and incremental best practices as the scarce resource. Built a portfolio site with an "Ask the site" assistant answering questions grounded in a curated corpus of my own work (https://wosotowsky.org/adam).

PROFESSIONAL EXPERIENCE

Principal Data Architect | Zimperium | 2021–2026

- Owned company-wide data and telemetry strategy for a mobile-security platform, leading the Data, Edge, and Application Analysis engineering organizations.
- Directed the pipelines feeding mobile threat detection models, from raw device and app telemetry through feature extraction to classifier output and customer-facing verdicts.
- Created an AI-powered regional mobile threat assessment portal with animated threat maps, industry-specific assessments, relevant regulations, and auto-generated sales decks.
- Architected real-time edge-delivery systems serving threat intelligence and model results to customer consoles with low latency across regions.
- Led a full datacenter uplift from physical Rackspace infrastructure to virtualized Oracle Cloud, improving reliability, elasticity, and the cost profile of analytic workloads.
- Chaired the Principal Engineering Committee, setting technical direction and arbitrating architecture decisions; served as the executive-facing authority on telemetry interpretation for Sales and key accounts.

Continuous role at what became McAfee — the following five entries are a single team, acquired four times (CipherTrust → Secure Computing → McAfee → Intel → McAfee spinout, 2003–2021), with an internal promotion at every transition.

Research Architect | McAfee | 2016–2021

- Set architectural direction for the messaging, connection-reputation, and malware research systems underpinning Global Threat Intelligence.
- Designed new reputation-scoring services delivering model output to firewall and email products at global scale.
- Built a unified research portal fusing disparate intelligence sources into one analyst workspace, cutting investigation and labeling time for the research team.
- Managed end-of-life of legacy GTI messaging systems while maintaining global reputation service stability.

Data Architect | Intel | 2012–2016

- Drove data strategy and classifier development for threat-intelligence platforms processing internet-scale traffic telemetry.
- Owned the quality and operation of ~200 production classifiers, with a hand in the design or

implementation of more than half: feature selection, training-data curation, tuning, and false-positive control.
- Personal research output of ~30M IP and 200K domain blocks per year, directly shaping global protection outcomes and customer-visible false-positive rates.
- Replaced purchased third-party feeds with intelligence generated internally across our own product telemetry, materially reducing recurring data costs.

Principal Engineer | McAfee | 2010–2012

- Maintained and extended the Messaging GTI classifier suite and the Hadoop extraction pipelines generating its training and scoring data.
- Unified workflow and content streams across multiple acquired email-security product lines into a coherent intelligence pipeline.
- Senior technical owner for top-tier customer escalations, coordinating resolution across research and engineering.

Special Agent | Secure Computing | 2006–2010

- Designed and produced the complete anti-spam configuration for the email appliance, including heuristic tuning, scoring thresholds, and end-to-end configuration management.
- Partnered with research on classifier design, translating field observations into detection features for emerging threats.
- Delivered zero-day fixes under customer-critical timelines; trained engineers on backend systems, UNIX internals, and shell-based data analysis.

L3 Support | CipherTrust | 2003–2006

- Final escalation point for the most complex customer issues; prioritized defect remediation and guided engineering on customer-impacting problems.
- Built the appliance's "agent" infrastructure — autonomous daemons on the backend that triaged bugs, generated custom customer reports, and delivered new features without manual intervention.

PATENTS

- **US 11,743,276** — Methods, systems, articles of manufacture and apparatus for producing generic IP reputation through cross-protocol analysis.
- **US 11,689,550** — Methods and apparatus to analyze network traffic for malicious activity.

SPEAKING, TEACHING & COMMUNITY

- 13+ conference presentations and 2 keynotes on emerging threat trends; 4 training courses delivered (M3AAWG, NCFTA-adjacent, Texas ISF), including courses on shell-based anomaly surfacing in logs and raw data.
- M3AAWG presenter on threat intelligence, telemetry, and leveraging third-party data to maximize machine learning and reputation quality; GDPR working-group contributor and FBI escalation contact.
- 9 years on the McAfee Public Relations team (7 on the steering committee); hundreds of press placements across CNN, ABC News, LA Times, IEEE, and international trade publications.

EDUCATION

B.S. Computer Science, Minor in Systems Engineering — Georgia Institute of Technology

CIVIC & VOLUNTEER

Election poll worker • Office emergency responder • Disaster management team • Event committee